



The Impact of Shipping Time on Customer Satisfaction

Statistical Analysis using Correlation and A/B Testing

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le wagon

Company & Business Problem

- THE COMPANY

Olist is a Brazilian retail technology company that provides an integrated ecosystem of solutions for merchants, connecting ecommerce, marketplaces, business management and logistics.

- BUSINESS PROBLEM

Customer satisfaction ratings showed considerable variation across orders, raising a key business question: could delivery performance be influencing the customer experience?

Hypothesis: Longer shipping times may be associated with lower customer satisfaction ratings.

How does shipping time relate to customer satisfaction?



DATA & ANALYTICAL APPROACH

To investigate whether delivery performance is associated with customer satisfaction, the analysis focuses on:



Shipping Time

Number of days
between purchase and
delivery



Delivery Status

On-time vs. late
deliveries



Review Score

Customer rating from
1 to 5

Data & Analytical approach:

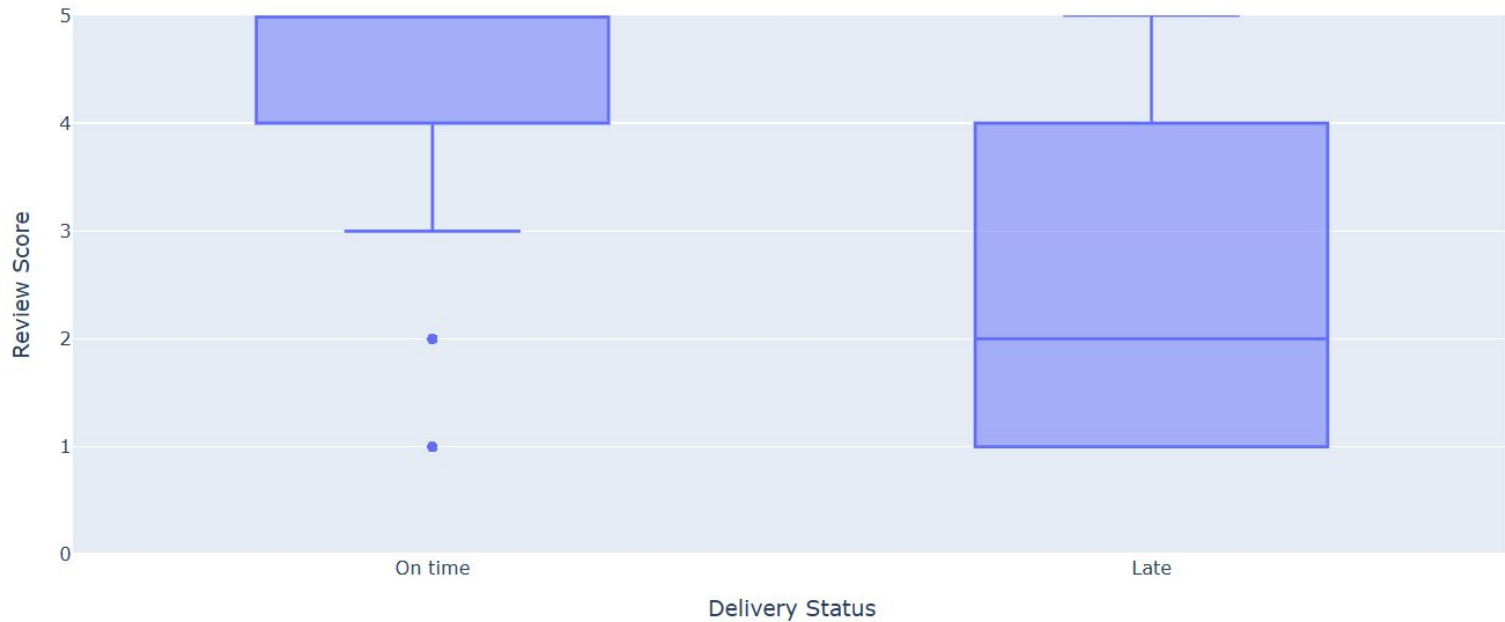
Descriptive Analysis → Correlation Analysis → Statistical Hypothesis Testing

Average Customer Ratings Are 1.64 Points Lower for Late Deliveries



Customer Ratings Show Three Times Greater Spread for Late Deliveries

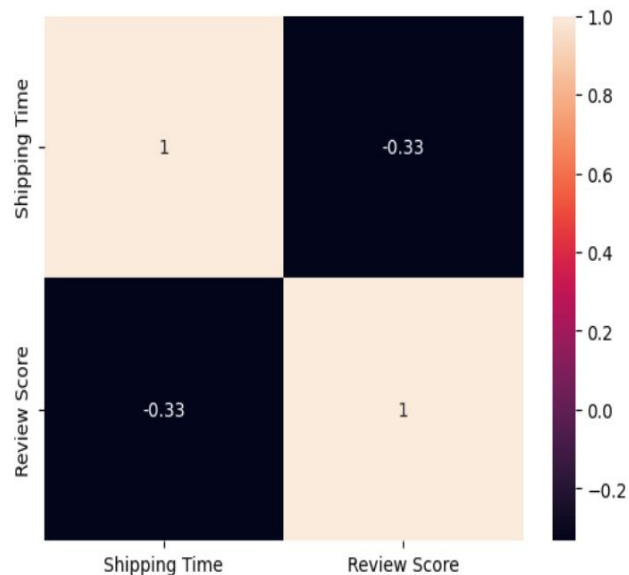
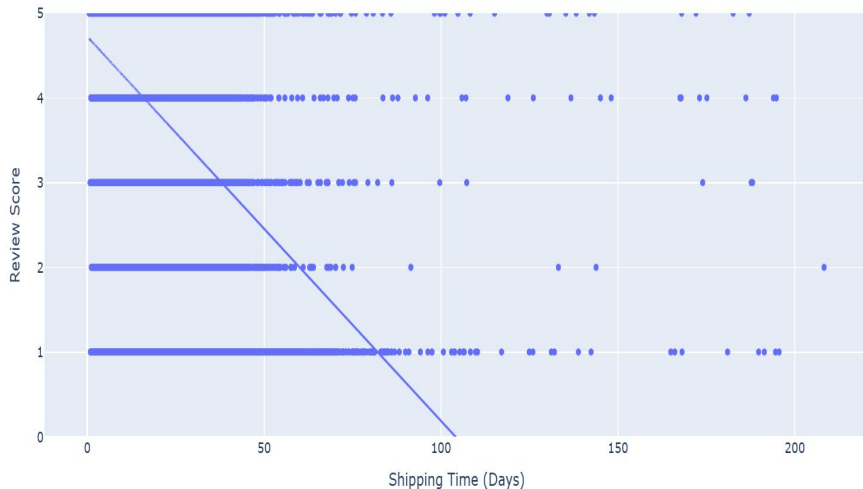
Review Score Distribution by Delivery Status



CORRELATION ANALYSIS

Longer Shipping Times Show Only a Moderate Negative Association with Customer Ratings

Shipping Time vs Review Score



Despite the negative association, the -0.33 correlation suggests that shipping duration alone does not necessarily determine customer ratings, raising a further question: **does delivery performance relative to the promised date matter more than shipping time itself?**

Delivery Status Is Significantly Associated with Customer Ratings



TEST

An independent two-sample t-test was conducted to determine whether average customer ratings differ significantly between on-time and late deliveries.

Do customer ratings differ systematically between on-time and late deliveries?

H₀: The mean customer rating is the same for on-time and late deliveries.

H₁: The mean customer rating is different for on-time and late deliveries.

On time: 4.21 ★ Late: 2.57 ★
t-statistic: 85.23 p-value: < 0.001

Result: Reject H₀

CONCLUSION

There is strong statistical evidence that average customer ratings differ between on-time and late deliveries.

```
print("Mean On time:", on_time.mean())
print("Mean Late:", late.mean())
print("T-statistic:", t_stat)
print("P-value:", p_value)

if p_value < 0.05:
    print("Reject H0")
    print("On-time deliveries have a significantly higher average review score.")
else:
    print("Fail to reject H0")
```

Last executed at 2026-08-27 20:24:01 in 4ms

Mean On time: 4.214306786272303
Mean Late: 2.566549798727438
T-statistic: 85.23240839395463
P-value: 0.0
Reject H0
On-time deliveries have a significantly higher average review score.

Business Insights & Recommendations



Key Findings

- On-time deliveries received **substantially higher** ratings: 4.21 vs. 2.57.
- Shipping time showed a **negative correlation** with customer ratings ($r \approx -0.33$).
- Ratings **differed significantly** between on-time and late deliveries ($p < 0.001$).



Business Recommendations

- Investigate delivery routes, regions and carriers with higher late-delivery rates.
- Use route-level analysis to identify and prioritize operational improvements.

Next Business Step: Analyze delivery performance by region, seller and carrier to identify key drivers of delays.



Thank You

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